

ENDOCRINE PHYSIOLOGY OBJECTIVES

ADRENAL CORTEX HORMONES.

1. Know the structural class of adrenal cortex hormones, the general class of hormone produced by each histological zone, and the predominant hormone secreted by each zone.
2. Know the basic properties of glucocorticoids and compare and contrast with other adrenal cortical hormones as well as protein and amino-acid derived hormones.
3. Be able to describe the physiological actions of glucocorticoids and its target tissues including actions on energy stores of muscle, fat and liver.
4. Know the net effects of epinephrine actions on plasma glucose, FFA, and amino acids.
5. Compare and contrast the metabolic actions of glucocorticoids with those of growth hormone, catecholamines, thyroid hormones, insulin and glucagon.
6. Describe the general processes related to inflammation that are affected by glucocorticoids that account for their anti-inflammatory effect?
7. Understand the role of glucocorticoids in the stress response and how it is distinguished from the fight or flight response mediated by catecholamines.
8. Define Cushing's disease and be able to classify this condition and other hyperglucocorticoid disorders based on clinical laboratory measurements of trophic and peripheral hormones.
9. Correlate clinical manifestations associated with Cushing's disease/syndrome with the normal physiological actions of glucocorticoids including development of adrenal diabetes.
10. Define Addison's disease and be able to classify this condition and other hypoglucocorticoid disorders based on clinical laboratory measurement of trophic and peripheral hormones.
11. Correlate clinical manifestations associated with Addison's disease with the normal physiological actions of glucocorticoids.
12. Know the basic properties of mineralcorticoids and compare and contrast with other adrenal cortical hormones as well as protein and amino-acid derived hormones.
13. Know the physiological actions of mineralcorticoids and its target tissues including the renal tubules.
14. Know the major regulatory mechanism that controls aldosterone secretion including the roles of renin and angiotensin II.
15. Know the pathophysiological manifestations developed from a mineralcorticoid deficiency and excess and be able to correlate with the normal physiological actions of mineralcorticoids.
16. Become familiar with interpreting the biochemical schematic outlining the biochemical biosynthesis of steroid hormones from cholesterol.
17. Understand why only certain steroids are produced in particular zones of the adrenal cortex.
18. Understand the overlapping functional nature of adrenal cortex steroids and the clinical implications of hyper- or hyposecretion.

19. Understand conceptually how deficient secretion of one P450 enzyme results in excess secretion of upstream intermediates and deficiency of hormone synthesis downstream from its action.
20. Be able to predict the steroid hormones that would be deficient and those would be in excess as a result of a given P450 enzyme deficiency (e.g., 21-hydroxylase deficiency)

ADRENAL MEDULLA HORMONES.

1. Know that catecholamines are biosynthesized for tyrosine by the adrenal medulla and that the major secretion product is epinephrine.
 2. Understand why a complete adrenalectomy would decrease the epinephrine to norepinephrine ratio in blood.
 3. Know the basic hormone properties of catecholamines and be able to compare and contrast to other amino acid derived hormones and to protein and steroid hormones.
 4. Be able to describe the physiological actions of adrenal catecholamines on energy metabolism including actions on muscle, fat, and liver energy stores.
 5. Know the net effects of epinephrine actions on plasma glucose, FFA, lactate, glycerol, and ketones.
 6. Compare and contrast the metabolic actions of adrenal catecholamines with those of growth hormone, glucocorticoids, thyroid hormones, insulin and glucagon.
 7. What are the general manifestations related to adrenal medulla insufficiency, sympathetic autonomic insufficiency? Medullary hypersecretion?
 8. Be able to correlate pathophysiological manifestations to the normal physiological actions of catecholamines.
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